Spar Design Optimization Under Loading Uncertainty Technical Report

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Executive Summary

This project focused on optimizing the design of a spar for a 500 kg aircraft that will be subject to stresses in a 2.5g maneuver with uncertainty in the load force distribution. Specifically, this report focuses on optimizing the spar mass to maximize fuel efficiency for a new long-endurance telecommunications UAV, subject to uncertainties in the expected aerodynamic loading.

The semi-wing span is 7.5 m, which is also the spar length. The spar shape is to be circular annulus shaped with varying inner and outer radii as the design variables. The inner radii are constrained to be within 0.01 m and 0.0475 m, and the outer radii constrained to be within 0.0125 m and 0.05 m. Both the inner and outer radii are constrained to be no less than 0.0025 m apart. Finally, no internal stresses within a 6 sigma safety margin (for the loading uncertainty) should exceed the ultimate strength of the carbon fiber composite material of the wing spar of 600 MPa. For the purposes of this study, the composite is modeled using effective isotropic properties.

The final optimized design was validated by a mesh convergence study and an optimization convergence study and was able to achieve a final mass of 8.574 kg, or about 70.758% lower than the 29.32 kg nominal design with a uniform inner and outer radii of 4.15 cm and 5 cm respectively.

Methodology

Analysis Method

Due to the symmetrical nature of the spar, this analysis considers one half of the spar's full length, with the spar's geometry being represented as a circular annulus shape. This geometry will be represented internally as a list of evenly spaced N nodes along the length of the spar, each with an inner and outer radii. With N nodes, the geometry is represented by a list of radii of length 2N. More specifically, the representation stores the radii in ascending order of node number, for example $[r_{i1}, r_{o1}, r_{i2}, r_{o2}, ...]$.

The geometry is constrained by the following constraints on the representational radii shown in the equations below.

$$0.01 \leq r_i \leq 0.0475 \quad (1)$$

$$0.0125 \leq r_o \leq 0.05 \quad (2)$$

$$r_{out} - r_{in} \geq 0.0025 \quad (3)$$

The objective of the optimization is to minimize the mass of the spar while maintaining all constraints, including ensuring the max stress is less than the ultimate strength of the material. To calculate the mass of a given design, the volume must first be calculated then multiplied by the density. To find the volume, the surface area is integrated from one node to the next. The mass of each segment can be solved by the equation below, and the total mass can be computed by summing the masses of each individual segment. Here the density, ρ, of the carbon fiber composite material used is $1,600\ kg/m^3$.

$$mass = \rho V = \rho \int_0^{\Delta X} \pi(r_o(x)^2 - r_i(x)^2)dx \quad (4)$$

$$total\,mass = \sum_{i=1}^{N} mass_i \quad (5)$$

The spar will be modeled using Euler-Bernoulli Beam Theory, with the assumptions of planar symmetry, cross-section varies smoothly, plane cross sections remain normal to longitudinal plane before and after bending, internal strain energy accounts only for bending moment deformations, deformations are small enough that nonlinear effects are negligible, and finally the material is assumed to be elastic and isotropic.

The displacement of the beam in the vertical direction is predicted by the 4th order PDE shown below.

$$\frac{d^2}{dx^2}(EI_{yy}\frac{d^2w}{dx^2}) = q \qquad \forall x \in [0,\ L] \quad (6)$$

Where 'w' is the vertical displacement in the z direction, 'q(x)' is the applied load per unit length, 'E' is the Young's modulus which is approximately 70 GPa for the chosen carbon fiber composite material, and 'I_{yy}' is the second-moment of inertia with respect to the y axis, and can be calculated for a circular annulus shape using the equation below.

$$I_{yy} = \pi\frac{D_o^4 - D_i^4}{64} \quad (7)$$

Once the Euler-Bernoulli equation is solved for w, these displacements can be used to solve for the magnitude of the normal stress as a function of x using the equation below, where 'z_{max}' is the maximum height of the cross section (or in other words, the outer radius).

$$\sigma_{xx}(x) = |\, z_{max} E \frac{d^2w}{dx^2}\, | \quad (8)$$

At the specified 2.5g maneuver, the force distribution in the spanwise direction has a linear distribution with maximum load at the root and zero load at the tip. The integral of the nominal loading must equal 2.5 times half the aircraft's weight, so the nominal loading is given by Equation (9) as seen below.

$$f_{nom}(x) = \frac{2.5W}{L}\left(1 - \frac{x}{L}\right) \tag{9}$$

However, in this case it is known that there is uncertainty in the loading and is modeled by Equation (10) below, where the probabilistic perturbation has the form seen in Equation (11) below, and the random variable ζ_n is sampled from the distribution $\zeta_n \sim N\left(0, \frac{f_{nom}(0)}{10n}\right)$.

$$f(x, \zeta) = f_{nom}(x) + \delta(x, \zeta) \tag{10}$$

$$\delta_f(x, \zeta) = \sum_{n=1}^{4} \zeta_n cos\left(\frac{(2n-1)\pi x}{2L}\right) \tag{11}$$

To apply this stress theory to the optimization problem, the Euler-Bernoulli beam equation is discretized using the finite-element method, where the solution is represented using hermite-cubic shape functions and the finite element equations result from the minimization of potential energy.

The MATLAB function 'fmincon' was used to perform the optimization of the design while enforcing the constraints. In order to enforce the geometry constraints of the spar, a matrix representation of linear inequalities was used to represent Equation (3) and passed into fmincon's 'A' and 'b' inequality arguments. To enforce the minimum and maximum bounds of the geometry, fmincon's lower and upper bound arguments were used to represent Equations (1 & 2).

Additionally, to account for the uncertainty in the force distribution and the strength of the material, our objective is to minimize the mass while also ensuring that the stress does not exceed some threshold such as the ultimate stress of the material. We can account for this uncertainty in the constraint seen below.

$$E[s, \zeta] + 6\sqrt{Var(s, \zeta)} \leq S_{ultimate} \tag{12}$$

To enforce the stress constraint and to ensure the spar can withstand the expected forces, fmincon's nonlinear constraint, 'nonlcon', argument was used. Here the constraint inequality is used to enforce the condition described below at each node. The Jacobian inequality is also computed by leveraging complex step differentiation in order to compute the partial derivatives of the stress constraint defined below.

$$c_i = \frac{E[s, \zeta] + 6\sqrt{Var(s, \zeta)}}{\sigma_{max}} - 1, \text{ such that } c_i \leq 0 \tag{13}$$

In order to use this constraint, we need to be able to calculate the mean and standard deviation of the stress distributions. To do this we will be leveraging Gauss-Hermite quadrature in order to accurately approximate the expected values which will be used to find the mean and standard deviations. Specifically we will use Equation (14) as shown below. To approximate it using the gauss-Hermite quadrature, we will need to transform the weights and locations using Equations (15 & 16).

$$E(\sigma) = \int_{-\inf}^{+\inf} \sigma(x, \zeta)P(\zeta, \mu, \sigma)\, d\zeta = \frac{1}{\sqrt{\pi}} \int_{-\inf}^{+\inf} f(\sqrt{2}\sigma u + \mu)e^{-u^2} du \tag{14}$$

$$w_i^* = \frac{1}{\sqrt{\pi}} w_i \tag{15}$$

$$u_i^* = \sqrt{2}\sigma\, u_i + \mu \tag{16}$$

Now that we have the weights and positions transformed for our use case, we can use them to approximate the expected values as seen in Equations (17 & 18). Finally we can use these values to calculate the standard deviation as seen in Equation (19). Note that when using these equations for our use case, we will actually need 4 nested summations since we need to do a 4d integration since we are dealing with 4 random variables, but the logic here is the same.

$$E[\sigma] = \mu = \sum_{i=1}^{m} w_i^* \sigma\left(x_{position}, u_i^*\right) \tag{17}$$

$$E[\sigma^2] = \sum_{i=1}^{m} w_i^* \sigma\left(x_{position}, u_i^*\right)^2 \tag{18}$$

$$\sigma_{std} = \sqrt{E[\sigma^2] - E[\sigma]^2} \tag{19}$$

Here $\sigma(x)$ is computed using the methods described earlier and is implemented in MATLAB by first computing the inertia at each node, leveraging the provided 'CalcBeamDisplacement' MATLAB function which estimates the beam displacements at each node using the methods described above, using the outer radii as z_{max}, and finally leveraging the provided 'CalcBeamStress' MATLAB function to compute the stresses in the beam at each node using the methods described above. All of the stress analysis described above was implemented in the 'spar_stress_analysis.m' file.

Finally, the 'nonlcon' constraint function is passed to 'fmincon' to enforce these nonlinear stress constraints at each node. When using 'fmincon' for the optimization process, several non-default settings were used. The `GradObj` and `GradConstr` flags are set to enable 'fmincon' to use analytical gradients and Jacobians supplied manually, potentially speeding convergence compared to numerical finite-differences. The `Display` set to `iter` shows progress at each step, while `Algorithm` set to `sqp` selects a sequential-quadratic-programming solver that handles nonlinear constraints efficiently.

Limitations

The discretization of the Euler-Bernoulli beam equation is an efficient way to evaluate any given spar design, however its discretization nature causes it to have some flaws that need to be addressed. The main flaw here is that with any discretization, you must allow some approximations. This problem is not discrete in nature, so a discretization of this problem is an approximation of the true solution. You would need an infinite amount of nodes to be perfectly accurate, however this is not feasible. Additionally, a

Gauss-Hermite quadrature, while highly accurate, is still an approximation, and its accuracy can depend on the number of points used.

To address these limitations, a full convergence study is performed to verify that the model can converge to the expected value provided a reasonable number of nodes/points. This will allow confidence that the design does indeed have properties similar to those predicted by the model analysis method.

Results

The objective for optimization is minimizing the mass of the spar design while subject to the constraints defined in methodology, such as the physical geometry/radii constraints, and the stress constraints including the uncertainties. This section covers a detailed analysis of the results of this optimization process including a nominal design, an initial design, and the optimized design.

Nominal Design

The initial design to what we will compare our final results to is a design with a uniform inner and outer radii of 4.15 cm and 5 cm respectively, where the total mass was found to be approximately 29.32 kg. The shape of this nominal design is shown below in Figure (1), which will serve as our benchmark to compare to.

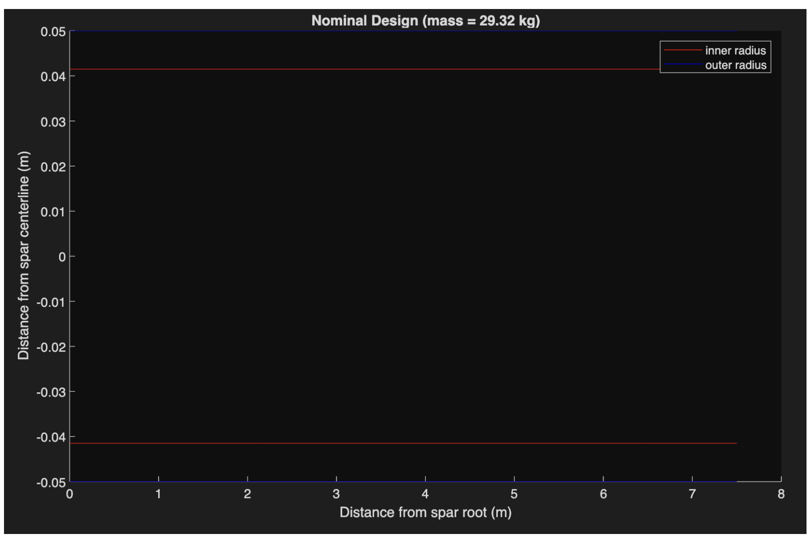

Figure 1 - Nominal spar design geometry (benchmark design)

To verify the number of quadrature points to be used in the stress analysis, we will plot the calculated stress at the center of the spar at varying numbers of quadrature points to ensure that it converges to some value so that we know that the approximation is representative of the actual values. As seen in Figure (2) below, the calculated mean and standard deviations begin to converge after ~3 quadrature points, validating the feasibility of the method. Due to this, for the remainder of the report we use 3 quadrature points to maintain high levels of accuracy in the analysis.

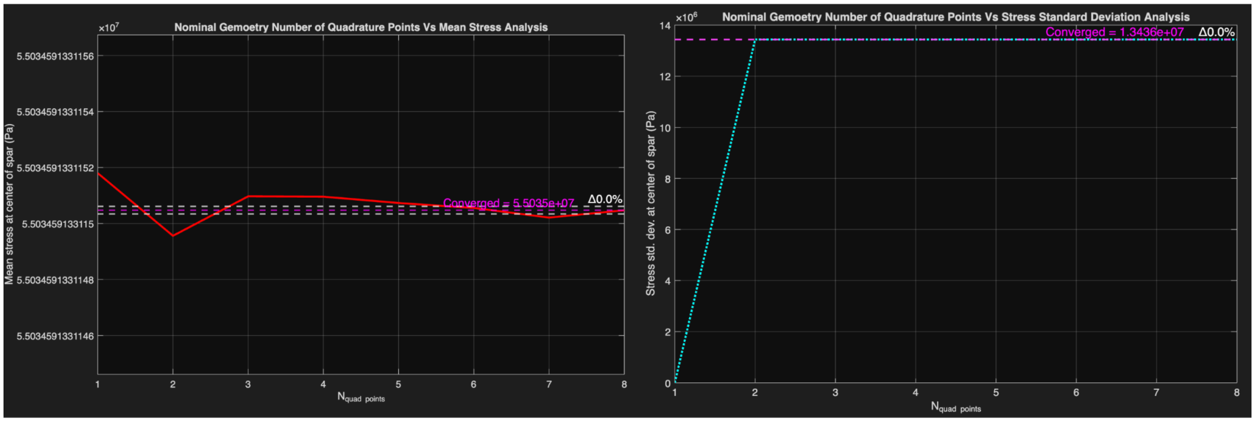

Figure 2 - Nominal design quadrature convergence study

To verify that the stress values are accurate from the stress analysis, a mesh convergence study was done on the nominal design analysing the stress at the end of the spar. The stress at the end of the spar is known to be 0 analytically. This can be used to verify if the geometry is of high enough fidelity to be useful, and if the numbers are representative of how the design would behave in the real world. Below Figure (3) shows the stress mean and standard deviation at the end of the spar at varying amounts of nodes. As seen in Figure (3), with higher amounts of nodes, the stress at the end of the spar approaches 0 Pa, meaning the numbers are representative of how they would behave in the real world. We can see that at around 15 nodes, the value converges, so for the remaining analysis we will use 15 Nodes to maintain accuracy.

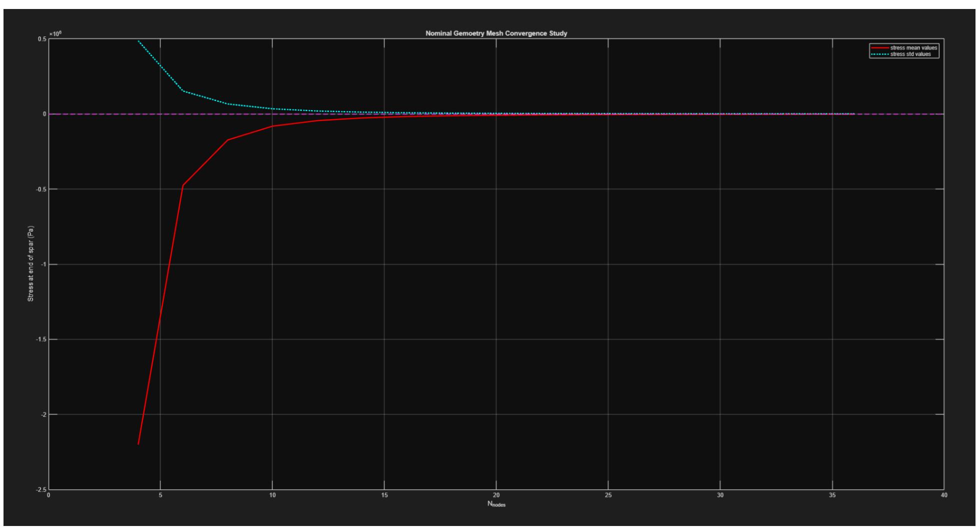

Figure 3 - Nominal spar design mesh convergence study

Finally after verifying the methods used in the analysis, to investigate the expected stress of the nominal design, the mean and standard deviation of the stress was plotted against the length of the spar as seen in Figure (4) below, and verified against a known distribution to ensure a correct implementation.

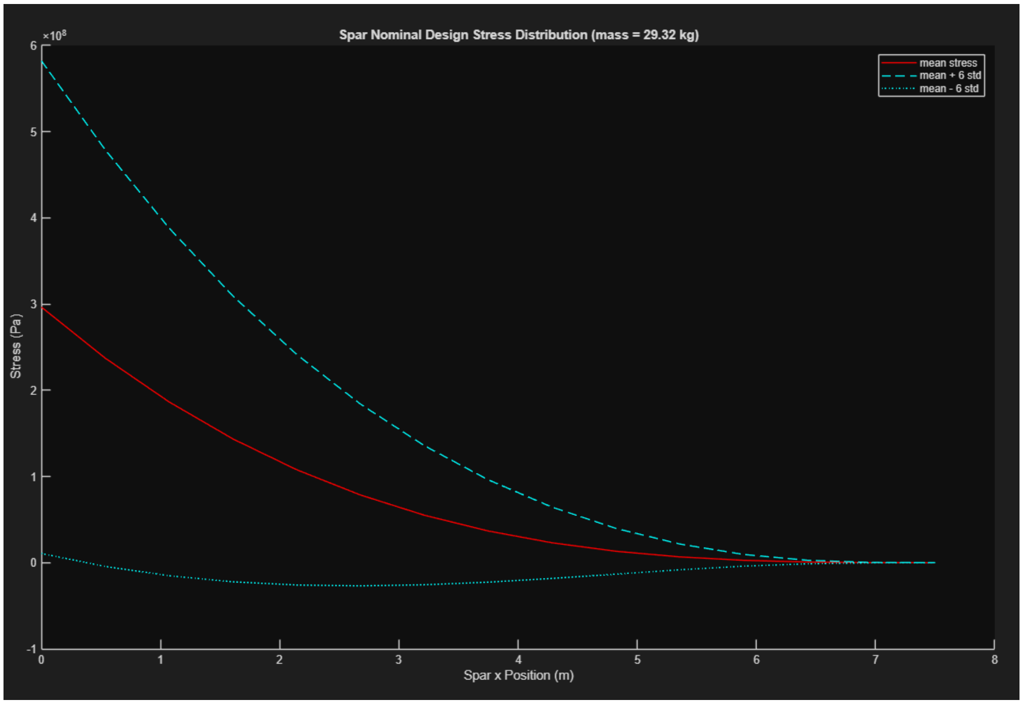

Figure 4 - Nominal spar design stress distribution

Initial Unoptimized Design

The initial design before optimization however, is defined by N=15 nodes (as verified earlier) such that it is initialized to have a uniform inner and outer radii of 0.015 m and 0.045 m respectively, leading to a uniform initial geometry as shown below in Figure (5), this will be used as our initial starting guess/state for the optimization.

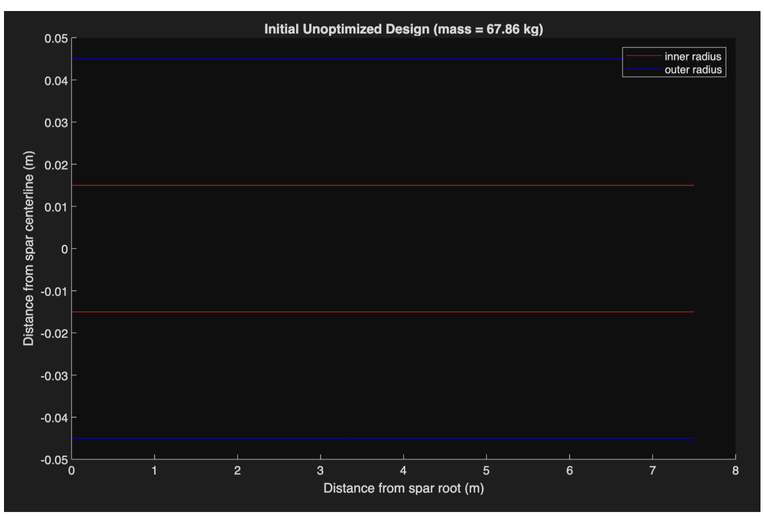

Figure 5 - Initial spar design geometry before optimization (initial optimization guess)

Final Optimized Design

After running the optimization process with the parameters initialized above, using MATLAB's 'fmincon' function, the algorithm landed on the geometry shown below as the most optimal geometry found given the constraints. With an estimated mass of 8.57 kg, or approximately 70.76% lower than the 29.32 kg nominal value. The final geometry can be seen in Figure (6) below, and the final optimal design parameters can be found in the appendix.

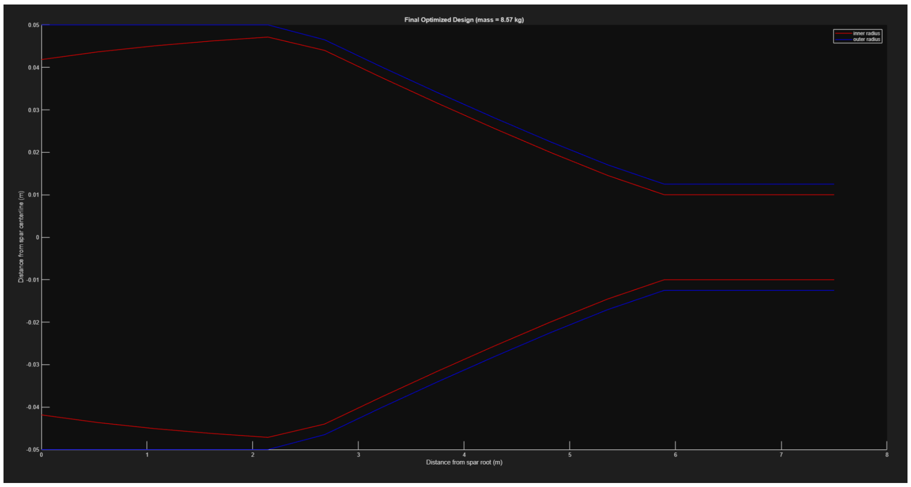

Figure 6 - Optimized spar design geometry

To show evidence that the 'fmincon' function has reached some sort of local minimum, below Figure (7) shows the optimization convergence plot which shows the first order optimality, feasibility, and the objective value all plotted against the optimization iteration number. In the plot we can see the first order optimality is decreasing as expected before decreasing to 0, thus showing that it is steadily approaching a local minimum, while at the same time, the feasibility steadily decreases to 0 with optimization steps meaning the design is adhering to the constraints defined in the optimization statement when it finally converges to 0 meaning the final design fully adheres to all constraints.

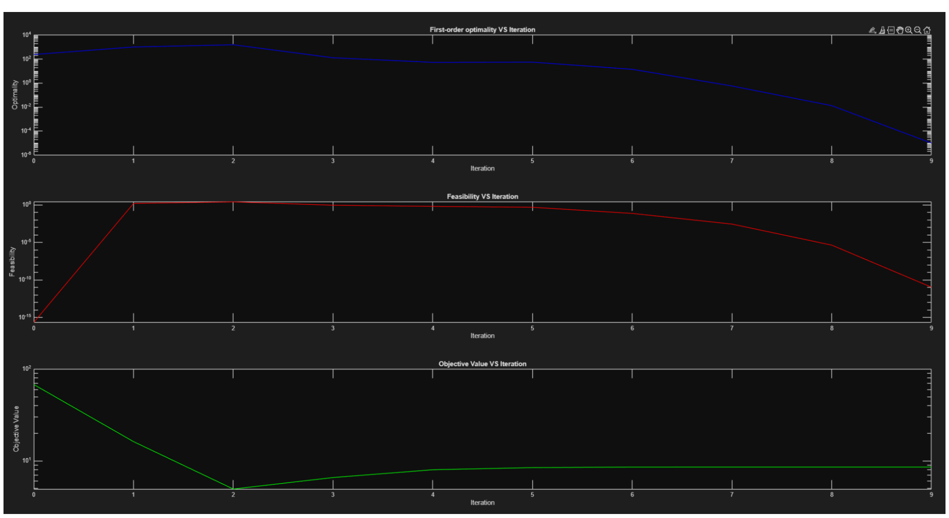

Figure 7 - Optimization convergence study

Finally, to analyze the expected stress distribution in the final optimized design, the mean stress and six times the standard deviation were plotted against the length of the spar as seen below in Figure (8). As seen below the final design adheres to the stress constraint at every location, such that the mean stress plus 6 times the stress standard deviation is always below the ultimate strength of the material (600 MPa).

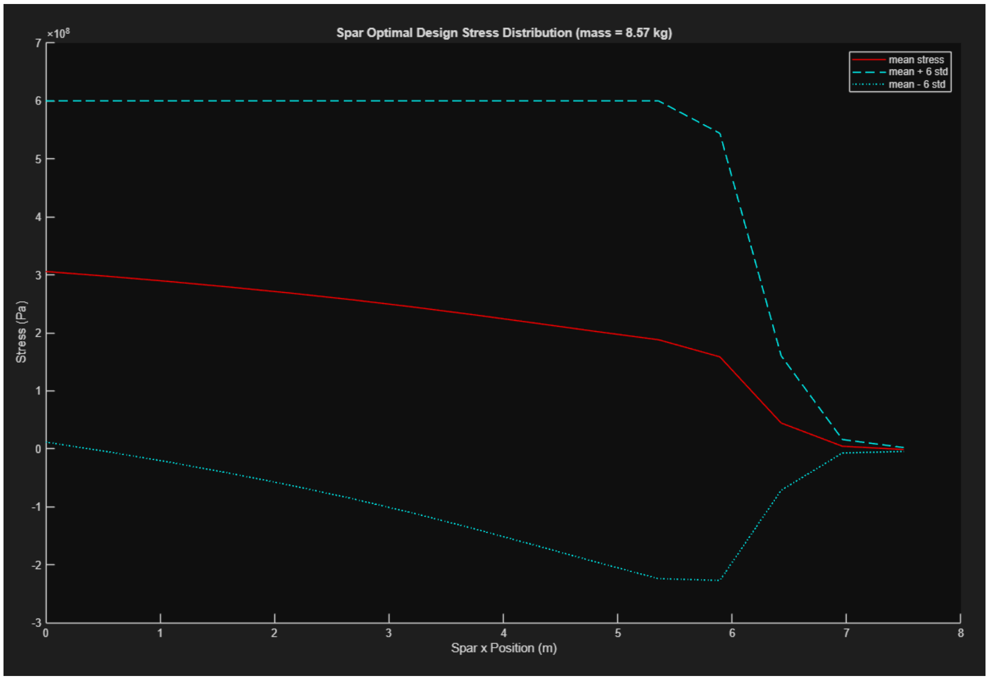

Figure 8 - Optimal spar design stress distribution

Conclusions

This project successfully optimized a wing spar design under uncertain loading, achieving a final mass of 8.57 kg which is 70.76% lighter than the 29.32 kg nominal design while satisfying all geometric constraints and the robust stress condition (mean stress + 6 standard deviations ≤ 600 MPa). The Gauss-Hermite quadrature method accurately quantifies stress statistics using 3-point integration per dimension, validated by the quadrature convergence studies. The 15-node mesh was also verified by the mesh convergence study to accurately quantify stresses. Finally the optimal design consistently met all constraints, with the stress distribution showing the 6σ safety margin, as seen in Figure (8), confirming reliability under load uncertainties.

This robust optimization showed a major reduction in final mass, while also ensuring safety against real-world load variations. This work highlights that uncertainty-aware methods are essential for aerospace applications, where balancing weight reduction with safety margins is critical for long-endurance UAVs. Future work could refine quadrature accuracy or explore additional load models to further enhance robustness.

Appendix

Parameters for Optimized geometry ($N_{nodes} = 15$)

[
0.041837237861469
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Citations

Hicken, Jason. Project #2 & 4 Model Notes.
Numerical Design Optimization, Rensselaer Polytechnic Institute, 16 Oct. 2025.
Slides provided by the professor.

Hicken, Jason. Project #2 & 4 Notes.
Numerical Design Optimization, Rensselaer Polytechnic Institute, 16 Oct. 2025.
Slides & constraint/beam stress FEA code provided by Dr. Hicken.